Enumerative Combinatorics

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This book is intended to be a problem sequence for a one-semester graduate enumerative combinatorics course that utilizes an inquiry-based learning (IBL) approach. You may not modify the content of this book without expressed consent of the author.

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The mathematician does not study pure mathematics because it is useful; he studies it because he delights in it, and he delights in it because it is beautiful.

Henri Poincaré, mathematician & physicist

Chapter 1

Introduction

1.1 What is This Course All About?

This course focuses on enumerative combinatorics with an introduction to generating function techniques.

This course will use an inquiry-based learning (IBL) to learning combinatorics. Rather than spending most of class listening to lectures, you will spend substantial time exploring problems, developing arguments, presenting solutions, and discussing the mathematics with your classmates.

Outside of class, you may use whatever resources help you learn—including books, websites, classmates, and AI tools such as ChatGPT. The important distinction is between **finding a solution** and **understanding the mathematics**. When you come to class, you should be prepared to explain, defend, critique, and adapt the ideas you encountered.

Because outside resources are readily available, homework and preparation will be relatively low-stakes. Your understanding will be assessed primarily through your mathematical contributions in class and regular short, individual, no-resource assessments. In short:

Use whatever helps you learn. Be prepared to demonstrate that you learned it.

1.2 Inquiry-Based Learning

Mathematics is not about calculations, but ideas. My goal as a teacher is to provide students with the opportunity to grapple with these ideas and to be immersed in the process of mathematical discovery. Repeatedly engaging in this process hones the mind and develops mental maturity marked by clear and rigorous thinking. Like music and art, mathematics provides an opportunity for enrichment, experiencing beauty, elegance, and aesthetic value. The medium of a painter is color and shape, whereas the medium of a mathematician is abstract thought. The creative aspect of mathematics is what captivates me and fuels my motivation to keep learning and exploring.

While the content we teach our students is important, it is not enough. An education must prepare individuals to ask and explore questions in contexts that do not yet exist and to be able to tackle problems they have never encountered. It is important that we put these issues front and center and place an explicit focus on students producing, rather than consuming, knowledge. If we truly want our students to be independent, inquisitive, and persistent, then we need to provide them with the means to acquire these skills. Their viability as a professional in the modern workforce depends on their ability to embrace this mindset.

When I started teaching, I mimicked the experiences I had as a student. Because it was all I knew, I lectured. By standard metrics, this seemed to work out just fine. Glowing student and peer evaluations, as well as reoccurring teaching awards, indicated that I was effectively doing my job. People consistently told me that I was an excellent teacher. However, two observations made me reconsider how well I was really doing. Namely, many of my students seemed to depend on me to be successful, and second, they retained only some of what I had taught them. In the words of Dylan Retsek:

"Things my students claim that I taught them masterfully, they don't know."

Inspired by a desire to address these concerns, I began transitioning away from direct instruction towards a more student-centered approach. The goals and philosophy behind inquiry-based learning (IBL) resonate deeply with my ideals, which is why I have embraced this paradigm. According to the Academy of Inquiry-Based Learning, IBL is a method of teaching that engages students in sense-making activities. Students are given tasks requiring them to solve problems, conjecture, experiment, explore, create, and communicate—all those wonderful skills and habits of mind that mathematicians engage in regularly.

In an IBL course, instructor and students have joint responsibility for the depth and progress of the course. While effective IBL courses come in a variety of forms, they all possess a few essential ingredients. According to Laursen and Rasmussen (2019), the Four Pillars of IBL are:

- Students engage deeply with coherent and meaningful mathematical tasks.
- Students collaboratively process mathematical ideas.
- Instructors inquire into student thinking.
- Instructors foster equity in their design and facilitation choices.

This book can only address the first pillar while it is the responsibility of your instructor and class to develop a culture that provides an adequate environment for the remaining pillars to take root.

Just like learning to play an instrument or sport, you will have to learn new skills and ideas. Along this journey, you should expect a cycle of victory and defeat, experiencing a full range of emotions. Sometimes you will feel exhilarated, other times you might be seemingly paralyzed by extreme confusion. You will experience struggle and failure

before you experience understanding. This is part of the normal learning process. If you are doing things well, you should be confused on a regular basis. Productive struggle and mistakes provide opportunities for growth. As the author of this text, I am here to guide and challenge you, but I cannot do the learning for you, just as a music teacher cannot move your fingers and your heart for you.

You could view this book as mountaineering guidebook. I have provided a list of mountains to summit, sometimes indicating which trailhead to start at or which trail to follow. There will always be multiple routes to top, some more challenging than others. Some summits you will attain quickly and easily, others might require a multi-day expedition. Oftentimes, your journey will be laced with false summits. Some summits will be obscured by clouds. Sometimes you will have to wait out a storm, perhaps turning around and attempting another route, or even attempting to summit on a different day after the weather has cleared. The strength, fitness, and endurance you gain along the way will allow you to take on more and more challenging, and often beautiful, terrain. Do not forget to take in the view from the top! The joy you feel from overcoming obstacles and reaching each summit under your own will and power has the potential to be life changing. But make no mistake, the journey is vastly more important than the destinations.

> Don't fear failure. Not failure, but low aim, is the crime. In great attempts it is glorious even to fail.
>
> Attributed to Bruce Lee, martial artist & actor

Chapter 2

Preliminaries

2.1 Combinatorial Structures

One of our objectives is to study combinatorial structures that arise in mathematics, with an eye towards enumeration and classification. We will begin by investigating several sets of discrete mathematical objects and exploring the connections among them. In no particular order, here are some combinatorial species[1] we may encounter throughout the semester:

Binary String	Pixel Strip	Lattice Path
Starfolks Path	Compositions	Pebble Piles
Domino Stacking	Domino Tiling	Permutation
Nonattacking Rooks	Sparse Strip	Odd Composition
Binary Tree	Increasing Tree	Triangulated Polygon
Ballot Path	Noncrossing Partition	Parking Function
Unlabeled Tree	Set Partition	Integer Partition
Dodger Rooks	Cayley Permutation	Fubini Arrangement
Fraenkel Tree	Knuth Rooks	Penrose Doodle

Note that I invented some of the names mentioned above. Figure 2.1 provides a few examples of individuals one can find among the species mentioned above.
Below are some questions we should ponder when exploring a new species:

- What patterns do we see?
- How do they "grow"?
- What is a sensible definition of the species? How does your definition tie in to the **index** n and the **rank** r?
- Can you provide a generic example with index $n = 10$? For your example, what is r?

[1] In this context, I am using the word "species" in a colloquial sense. Later in the semester, this word will have a more precise mathematical meaning.

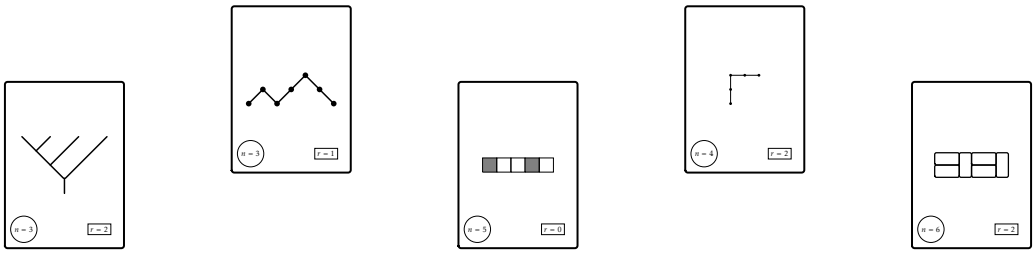


Figure 2.1: Examples of combinatorial species.

- How many objects are there of index n? Is there a nice formula?
- How many objects are there of index n and rank r? Is there a nice formula?
- For a fixed index n how do you generate all the possibilities? What is the simplest/most efficient way to generate them?
- How does this species relate to other species we have studied?

You may not be able to answer some of these questions immediately, and that's okay. The goals here are to have a variety of examples at our finger tips and to start a taxonomy of combinatorial species.

Problem 2.1. For each of the species you encountered or were assigned, complete the following.

(a) List the name of the species.

(b) Provide a working definition for the species. Your definition should include mention of the index n and the rank r, including the domain for n (and possibly r). For example, your definition might start as, "A *dinglehopper* of index $n \in \mathbb{N} \cup \{0\}$ and rank r is…"

(c) Provide a generic example for index $n = 10$. For your example, specify the rank r. Your example should illustrate most or all of the key features and not be too "boring."

(d) Provide a table of values that enumerates the number of structues of index n for small n. Your table should start at either $n = 0$ or $n = 1$ depending on the species and it should look something like the following:

n	0	1	2	3	4	5
count	1	1	2	3	5	8

(e) If you can, write down a formula (either a closed form or recursive) that counts the number of structures of index n. Justify your answer. Please do not look this up or ask AI. If you cannot do it at this time, I would prefer we leave it blank!

(f) Now, provide a table of values that enumerates the number of structures of index n and rank r for small values of n and r. Your table should start at either $n = 0$ or $n = 1$ and $r = 0$ or $r = 1$ depending on the species, with n labeling the rows and r labeling the columns. Table 2.1 illustrates the requested format.

$n\backslash r$	0	1	2	3	4	5
0	1					
1	1	1				
2	1	2	1			
3	1	3	3	1		
4	1	4	6	4	1	
5	1	5	10	10	5	1

Table 2.1: Sample counts of structures organized by index n and rank r.

(g) Ignoring the row labels, what happens if you sum the enteries in a row of your table in part (f)?

(h) If you can, write down a formula (either a closed form or recursive) that counts the number of structures of index n and rank r. Justify your answer. Please do not look this up or ask AI. If you cannot do it at this time, I would prefer we leave it blank!

(i) Are there any other species you have already encountered that seems related to this species?

2.2 First Principles

Let's establish some notation and make sure we are on the same page regarding some basic counting principles.

In this book, the set of **natural numbers** is given by $\mathbb{N} := \{1, 2, 3, \ldots\}$. Sometimes mathematicians include 0 in the natural numbers, but that will not be our convention. For $n \in \mathbb{N} \cup \{0\}$, the following notation is useful:

$$[n] := \begin{cases} \{1, 2, \ldots, n\}, & \text{if } n \in \mathbb{N} \\ \varnothing, & \text{if } n = 0. \end{cases}$$

The **cardinality** of a set A, denoted by $|A|$, is the number of elements in A. If A and B are sets, the **Cartesian product** of A and B, denoted $A \times B$ (read as "A times B" or "A cross B"), is the set of all **ordered pairs** where the first component is from A and the second component is from B. In set-builder notation, we have

$$A \times B := \{(a, b) \mid a \in A, b \in B\}.$$

Cartesian products are named after French philosopher and mathematician René Descartes (1596–1650).

Problem 2.2. If A is a set, then what is $A \times \varnothing$ equal to?

Problem 2.3. Under what circumstances is $A \times B = B \times A$?

The Cartesian product of sets $A_1, \ldots, A_n$ is the collection of n-**tuples** given by

$$A_1 \times \cdots \times A_n := \{(a_1, \ldots, a_n) \mid a_j \in A_j \text{ for all } 1 \leq j \leq n\},$$

where A_i is referred to as the ith **factor** of the Cartesian product. As a special case, the set

$$\underbrace{A \times \cdots \times A}_{n \text{ factors}}$$

is often abbreviated as A^n.

Theorem 2.4 (Product Principle)**.** Each of the following are referred to as the **Product Principle**.

(a) If $A_1, \ldots, A_k$ are finite sets, then

$$|A_1 \times \cdots \times A_k| = |A_1| \cdots |A_k|.$$

(b) If $\mathcal{O}$ is the set of outcomes for a k-step process, where for $1 \leq i \leq k$, there are n_i choices for step i, no matter what earlier choices were made, then

$$|\mathcal{O}| = n_1 n_2 \cdots n_k.$$

The key difference between the two versions of the Product Principle is that the second version does not assume that the set of choices for step i is independent of the previous choices.

Problem 2.5. Is your answer for Problem 2.2 compatible with the Product Principle?

For $n \in \mathbb{N} \cup \{0\}$, a **string** of objects of **length** n is a linearly ordered arrangements of n objects. The objects in a string may be repeated unless specified otherwise. A string of length 0 is referred to as the **empty string**. If a string consists of letters, then it may be called a **word** (we do not require a word to be in the dictionary!). For example, $NNENEEE$ is a word of length 7 consisting of N's and E's.

For $n \in \mathbb{N} \cup \{0\}$, a **binary string** is a string consisting of 0's and 1's. Each occurrence of a 0 or 1 in a binary string is called a **bit**. The strings 010 and 100 are two different binary strings of length 3.

Problem 2.6. For $n \in \mathbb{N} \cup \{0\}$, how many binary strings of length n are there?

Problem 2.7. For $n \in \mathbb{N} \cup \{0\}$, how many subsets does a set with n elements have? You should be able to carefully justify your answer by cleverly utilizing the Product Principle.

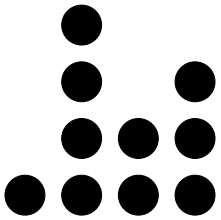


Figure 2.2: A pebble pile configuration consisting of 10 pebbles and 4 piles.

For $n \in \mathbb{N}$, a **pebble pile configuration** of index n and rank r consists of n indistinguishable dots arranged into r nonempty vertical piles. Each pile is bottom-justified along a common horizontal baseline, with the dots in a pile stacked vertically. The piles are arranged from left to right, and their order matters. Figure 2.2 depicts a pebble pile configuration with 10 pebbles and 4 piles.

Problem 2.8. For $n \in \mathbb{N}$, how many pebble pile configurations are there consisting of n pebbles?

For $n \in \mathbb{N}$ and $1 \leq r \leq n$, a **composition** of n with r parts is an ordered list of r positive integers whose sum is n. For example, $(1, 2, 1, 3)$ is a composition of 7 with 4 parts. An arbitrary composition with r parts may be denoted as $\alpha = (\alpha_1, \ldots, \alpha_r)$.

Problem 2.9. For $n \in \mathbb{N}$, how many compositions of n are there? Is there an obvious connection between pebble pile configurations and compositions? Can you explicitly describe this connection?

Problem 2.10. Imagine we have $n \geq 1$ distinct balls labeled $1, 2, \ldots, n$ and $k \geq 1$ distinct buckets labeled $1, 2, \ldots, k$. How many ways can we distribute the n distinct balls into the k distinct buckets?

Problem 2.11. Let $n, k \in \mathbb{N}$.

(a) How many functions are there from $[n]$ to $[k]$?

(b) If $k \leq n$, how many injective functions are there from $[k]$ to $[n]$?

(c) How many bijective functions are there from $[n]$ to $[n]$?

The game of chess is played on an 8×8 grid. We will consider chessboards with arbitrary dimensions, say $n \times k$ (n rows, k columns) with $1 \leq k \leq n$. A rook is a castle-shaped piece that can move horizontally or vertically any number of squares. In a typical chess game, there are two black rooks and two white rooks, but for our purposes we will assume we have k rooks, where all the rooks are the same color. We say that an arrangement of rooks on an $n \times k$ board is **nonattacking** if no two rooks lie in the same row or column. Figure 2.3 shows one possible nonattacking rook arrangement on a 4×4 board.

Problem 2.12. For $n \in \mathbb{N}$ and $1 \leq k \leq n$, how many nonattacking rook arrangements are there on an $n \times k$ board? What about an $n \times n$ board?

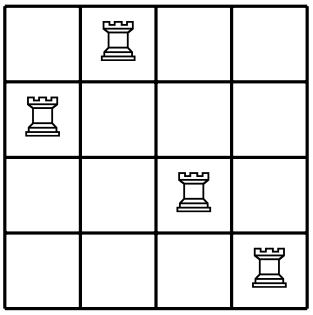


Figure 2.3: A nonattacking rook arrangement on a 4×4 board.

Theorem 2.13 (Sum Principle)**.** If $A_1, \ldots, A_n$ are pairwise disjoint sets, then

$$|A_1 \cup \cdots \cup A_n| = |A_1| + \cdots + |A_n|.$$

The Sum Principle is especially useful when a collection of objects can be partitioned into disjoint cases: count the objects in each case, then add the resulting counts.

Loosely speaking, a sequence of numbers is defined recursively if the nth term of the sequence is defined in terms of "earlier" terms of the sequence. One famous example of a sequence being defined recursively is the **Fibonacci sequence** (f_n), which we define by $f_0 = 1$, $f_1 = 1$, and $f_n = f_{n-1} + f_{n-2}$ for $n \geq 2$. The equation $f_n = f_{n-1} + f_{n-2}$ is the **recurrence relation** while $f_0 = 1$ and $f_1 = 1$ are the **initial conditions**. The first few terms of the Fibonacci sequence are:

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$$

It is important to emphasize that we cannot define the Fibonacci numbers using only the recurrence relation since otherwise, we would not be able to "get started" with the recurrence.

Problem 2.14. How many compositions α of n have the following properties?

(a) α has parts of size 1 and 2 only.

(b) α has only odd parts.

Problem 2.15. For $n \in \mathbb{N} \cup \{0\}$, how many binary strings of length n never have two consecutive 1's?

Problem 2.16. For $n \in \mathbb{N}$, consider a $2 \times n$ grid of squares. Using n dominoes with dimensions 1×2 squares, how many ways can we tile (i.e., perfectly cover with no overlap) the grid?

If a finite collection of sets is not pairwise disjoint, then we can still enumerate their union using the **Principle of Inclusion and Exclusion**.

Theorem 2.17 (Principal of Inclusion and Exclusion)**.** The number of elements in the union of finite sets $A_1, A_2, \ldots, A_n$ is

$$|A_1 \cup \cdots \cup A_n| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} |A_1 \cap A_2 \cdots \cap A_n|.$$

Problem 2.18. To ensure you understand the formula in the previous theorem, write down formulas for computing $|A \cup B|$ and $|A \cup B \cup C|$.

The next result, sometimes referred to as the **Subtraction Principle**, is simply a special case of Sum Principle.

Theorem 2.19 (Subtraction Principle)**.** If $A \subseteq U$, then $|A| = |U| - |A^c|$.

There are two use cases for the Subtraction Principle. Sometimes it might be easier to count the opposite of what you are looking to count and then to subtract from the size of the universe of discourse. The second approach involves "inflating" the universe of discourse to include additional elements, counting the larger set, and then subtracting off the extra stuff. The two approaches are really just two different perspectives of the same idea.

The **Division Principle** provides a method for ignoring "unimportant" differences when counting things. Loosely speaking, the general idea is to count a larger set of distinct objects and then use the Division Principle to "merge" the ones that are not significantly different from each other.

Theorem 2.20 (Division Principle)**.** If a finite set A is the union of n pairwise disjoint subsets each with d elements, then $|A| = n \cdot d$. Equivalently, $n = |A|/d$.

Figure 2.4 illustrates the Division Principle, where the set A contains 12 elements that are partitioned into $n = 3$ pairwise disjoint subsets, each of size $d = 4$. In this case, we see that $|A|/d = 12/4 = 3 = n$.

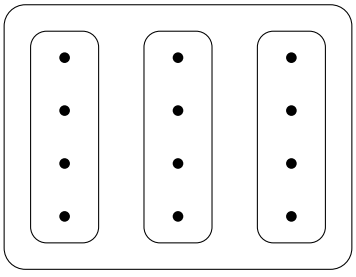


Figure 2.4: Visualization of the Division Principle with $|A| = 12$, $n = 3$, and $d = 4$.

Problem 2.21. How many ways can the letters of the word FLAGSTAFF be arranged?

Problem 2.22. Six friends sit around a circlular table to play a game. Rotations of the group do not constitute different seating orders.

(a) How many circular seating arrangements are there?

(b) How many circular seating arrangements are there if Sally and Maria always sit next to each other?

For $n \in \mathbb{N} \cup \{0\}$, a **set partition** of $[n]$ is a collection of nonempty pairwise disjoint subsets of $[n]$ whose union is $[n]$. Each subset in a set partition is called a **block**. For example, there are 7 set partitions of $[4]$ with 2 blocks, namely:

$$\begin{array}{ll} \{\{1,2,3\},\{4\}\} & \{\{1,2,4\},\{3\}\} \\ \{\{1,3,4\},\{2\}\} & \{\{2,3,4\},\{1\}\} \\ \{\{1,2\},\{3,4\}\} & \{\{1,3\},\{2,4\}\} \\ \{\{1,4\},\{2,3\}\} & \end{array}$$

The empty set has exactly one set partition, namely the empty collection of blocks, $\varnothing$. This partition has zero blocks; it is not $\{\varnothing\}$, since blocks must be nonempty.
We define the **Stirling numbers** (of the second kind) via

$$\left\{ {n \atop k} \right\} := \text{number of set partitions of } [n] \text{ with } k \text{ blocks.}$$

I usually pronounce this as "n Stirling k". Based on the information above, we know $\left\{ {4 \atop 2} \right\} = 7$. For $n \in \mathbb{N} \cup \{0\}$, we define the nth **Bell number** B_n to be the total number of set partitions of $[n]$ into any number of blocks. In particular, $B_0 = 1$, since the empty set has exactly one set partition.

The next problem is quite involved but provides an opportunity to put all of the counting principles to use.

Problem 2.23. Answer each of the following.

(a) Explain why $\left\{ {n \atop 1} \right\} = 1 = \left\{ {n \atop n} \right\}$ for every $n \in \mathbb{N}$.

(b) Explain why $\left\{ {n \atop 2} \right\} = 2^{n-1} - 1$ for every $n \in \mathbb{N}$.

(c) Explain why the Stirling numbers satisfy the following recurrence for $1 \leq k \leq n$:

$$\left\{ {n \atop k} \right\} = \left\{ {n-1 \atop k-1} \right\} + k \left\{ {n-1 \atop k} \right\}.$$

(d) Compute the Bell number B_4 by hand by writing down all of the appropriate set partitions.

(e) For $n \geq 1$, explain why

$$B_n = \sum_{k=1}^{n} \left\{ n \atop k \right\}.$$

Problem 2.24. Notice that back in Problem 2.11 we avoided counting surjective functions. We now have the tools to at least describe the count. Let $n, k \in \mathbb{N}$ wih $k \leq n$. How many surjective functions are there from $[n]$ to $[k]$?

Problem 2.25. Imagine we have $n \in \mathbb{N}$ people competing in a contest where ties are allowed.

(a) Explain why the number of final rankings is given by

$$\sum_{k=1}^{n} k! \left\{ n \atop k \right\}.$$

(b) Now, imagine that the people are numbered 1 through n. How many final rankings are possible under the proviso that person i's ranking is less than or equal to person $(i+1)$'s ranking? Such rankings are called **weakly increasing**.

2.3 Bijections

The next theorem states an important counting technique, which we refer to as the **Bijection Principle**.

Theorem 2.26 (Bijection Principle)**.** If there exists a bijection between finite sets A and B, then $|A| = |B|$.

For $n \in \mathbb{N} \cup \{0\}$, a **lattice path** of length n is a path in the integer lattice that begins at the origin and consists of n steps, each of which moves one unit north or one unit east. A lattice path of length n with r north steps is said to have rank r. Figure 2.5 provides an example of a lattice path of length 5 with rank 2.

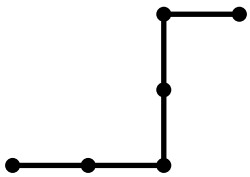


Figure 2.5: A lattice path of length 5 with rank 2.

For $n \in \mathbb{N}$ and $0 \leq r \leq n-1$, a **sticks and stones configuration** with n stones and r sticks is a string of n $\circ$'s and r |'s such that every | is framed by two $\circ$'s. For example, $\circ \mid \circ\circ\circ \mid \circ\circ$ is a sticks and stones configuration with 6 stones and 2 sticks.



Figure 2.6: A sparce strip of length 5 with rank 2.

For $n \in \mathbb{N} \cup \{0\}$, a **sparce strip** of length n is a $1 \times n$ grid such that each square in the grid is colored white or gray with the proviso that two gray squares are never adjacent. A sparce strip with r gray squares is said to have rank r. Figure 2.6 provides an example of a sparce strip of length 5 with rank 2.

Problem 2.27. Describe bijections between the following pairs of collections.

(a) For $n \in \mathbb{N} \cup \{0\}$, the collection of subsets of $[n]$ and the collection of binary strings of length n.

(b) For $n \in \mathbb{N} \cup \{0\}$, the collection of subsets of $[n]$ and the collection of lattice paths of length n.

(c) For $n \in \mathbb{N}$, the collection of compositions of n and the collection of sticks and stones configurations with n stones.

(d) For $n \in \mathbb{N}$, the collection of pebbles piles with n pebbles and the collection of sticks and stones configurations with n stones.

(e) For $n \in \mathbb{N} \cup \{0\}$, the collection of pebble pile configurations with $n + 1$ pebbles and the collection of subsets of $[n]$.

(f) For $n \in \mathbb{N}$, the collection of nonattacking rook arrangements on an $n \times n$ board and the collection of bijections from $[n]$ to $[n]$.

(g) For $n \in \mathbb{N}$, the collection of outcomes of a contest with n people with ties allowed (see part (a) of Problem 2.25) and the collection of surjective functions with domain $[n]$.

(h) For $n \in \mathbb{N}$, the collection of outcomes of a contest with n people as described in part (b) of Problem 2.25 and the collection of sticks and stones configurations with n stones.

(i) For $n \in \mathbb{N}$, the collection of compositions of n with parts of size 1 and 2 only (see Problem 2.14) and the collection of domino tilings of a $2 \times n$ grid (see Problem 2.16).

(j) For $n \in \mathbb{N}$, the collection of domino tilings of a $2 \times n$ grid and the collection of sparce strips of length $n - 1$.

In part (a) of Problem 2.27 you likely described a bijection between the collection of subsets of $[n]$ and the collection of binary strings of length n that also respected some additional structure. In particular, your bijection probably mapped k-subsets to binary strings with k ones. This is an example of a "nice" bijection, i.e., a set "isomorphism" that preserves combinatorial structure. In combinatorics, we really like these kinds of bijections.

Remark. Throughout these notes, we sometimes identify combinatorial objects with simpler representations when a specified bijection allows us to recover either uniquely from the other. Under this convention, we may say that an object "is" its representation, rather than repeatedly saying that it "corresponds uniquely to" that representation. For example, we may identify a nested ordered pair $(a,(b,c)) \in A \times (B \times C)$ with the ordered triple $(a,b,c) \in A \times B \times C$. These are different formal objects, but each uniquely determines the other. Such identifications may suppress braces, commas, or other notation, but they must retain enough information to recover the original object.

Problem 2.28. Using the Bijection Principle together with the previous problem and problems that appeared earlier, enumerate the following collections.

(a) For $n \in \mathbb{N} \cup \{0\}$, lattice paths of length n.

(b) For $n \in \mathbb{N}$, sticks and stones configurations with n stones.

(c) For $n \in \mathbb{N}$, the collection of surjective functions with domain $[n]$.

(d) For $n \in \mathbb{N} \cup \{0\}$, sparce strips of length n.

2.4 Combinations and Permutations

If A is a set and $B \subseteq A$ with $|B| = k$, we refer to B as a k-**subset** of A. The collection of all k-subsets of A is defined via

$$\binom{A}{k} := \{B \subseteq A \mid |B| = k\}.$$

For $n, k \in \mathbb{Z}$, the **binomial coefficients** are defined via

$$\binom{n}{k} := \begin{cases} \text{number of } k\text{-subsets of an } n\text{-element set}, & \text{if } 0 \leq k \leq n \\ 0, & \text{otherwise.} \end{cases}$$

In particular, if $|A| = n$, then $|\binom{A}{k}| = \binom{n}{k}$. We read "$\binom{n}{k}$" as "$n$ choose k". Alternate notations for binomial coefficients include $C(n,k)$ and ${}_nC_k$. We will see later why $\binom{n}{k}$ is referred to as a binomial coefficient.

Example 2.29. If $A = \{a,b,c,d\}$, then

$$\binom{A}{2} = \{\{a,b\},\{a,c\},\{a,d\},\{b,c\},\{b,d\},\{c,d\}\},$$

which implies that $\binom{4}{2} = 6$.

Problem 2.30. Let A be a finite set with $|A| = n \in \mathbb{N} \cup \{0\}$.

(a) What is $\binom{A}{0}$?

(b) What is $\binom{n}{0}$? Where do these values appear in Pascal's Triangle?

(c) What is $\binom{A}{n}$?

(d) What is $\binom{n}{n}$? Where do these values appear in Pascal's Triangle?

If we let n and k vary, we can organize the (nonzero) binomial coefficients in a triangular array, often referred to as **Pascal's Triangle**. See Table 2.2.

$n\backslash k$	0	1	2	3	4	5	6
0	1						
1	1	1					
2	1	2	1				
3	1	3	3	1			
4	1	4	6	4	1		
5	1	5	10	10	5	1	
6	1	6	15	20	15	6	1

Table 2.2: Pascal's Triangle of binomial coefficients.

Problem 2.31. Explain why the row sums in Pascal's Triangle are powers of two. In particular, for $n \geq 0$, explain why

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

We say that the binomial coefficients $\binom{n}{k}$ are a **refinement** of 2^n. This illustrates a general phenomenon we will witness over and over again.

Problem 2.32. Using the definition in terms of k-subsets (as opposed to some formula you might know involving factorials) together with a bijection, prove that for $0 \leq k \leq n$,

$$\binom{n}{k} = \binom{n}{n-k}.$$

This result is sometimes referred to as the **symmetry identity for binomial coefficients** (or simply **binomial symmetry**). Can you explain why it has this name?

The next result is called **Pascal's Recurrence** (or **Pascal's Identity**).

Problem 2.33 (Pascal's Recurrence)**.** Using the definition in terms of k-subsets, prove that for $n \in \mathbb{N}$ and $0 \leq k \leq n$,

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

The next problem illustrates a powerful idea: If we count the same collection in two different ways, then the two answers must agree.

Problem 2.34. Explain why

$$1 + 2 + \cdots + n = \binom{n+1}{2}$$

by counting the same collection in two different ways.

For $n \in \mathbb{N}$, the number defined by $T_n := 1 + 2 + \cdots + n$ is called the nth **triangular number** (due to the shape we get by representing each number in the sum by a vertical stack of balls). We also set $T_0 = 0$ by convention.

Problem 2.35. For $n \in \mathbb{N}$ and $1 \leq k \leq n$, prove that the number of compositions of n with k parts is $\binom{n-1}{k-1}$.

For $k \in \mathbb{N}$ and a nonempty set A, a k-**permutation** of A is an injective function $w : [k] \to A$. The set of all k-permutations of A is denoted by $S_{A,k}$. If A happens to be the set $[n]$, we use the notation $S_{n,k}$. And if $n = k$, we write $S_n := S_{n,n}$ and refer to each n-permutation in S_n as a **permutation**. It's not coincidence that this notation coincides with the notation for the symmetric group that one encounters in abstract algebra. Define

$$P(n,k) := |S_{n,k}|.$$

By convention, we set $P(n,0) := 1$, including the case when $n = 0$.

Problem 2.36. For $1 \leq k \leq n$, describe a bijection between $S_{n,k}$ and the collection of nonattacking rook arrangments on a $n \times k$ board.

In light of Problems 2.12 and 2.36, we can conclude that

$$P(n,k) = n \cdot (n-1) \cdots (n+1-k) = \frac{n!}{(n-k)!}.$$

Problem 2.37. For $0 \leq k \leq n$, prove that

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Problem 2.38. I'm in the mood for a cup of coffee! Let's say that the nearest place to get a cup of coffee is four blocks East and three blocks North. Assuming I only walk East or North, how many different routes can I take to get there?

Problem 2.39. How many binary strings of length 6 have an odd number of 0's?

Problem 2.40. Show that the Fibonacci numbers satisfy the following identity for $n \geq 0$:

$$f_n = \sum_{k \geq 0} \binom{n-k}{k}.$$

Hint: There are a few natural approaches. One elegant approach would be to utilize a combinatorial argument with one of the collections of compositions in Problem 2.14.

What is the previous result saying in terms of Pascal's Triangle?

Problem 2.41. Recall the definition of the Bell numbers (where $B_0 = 1$). Prove that for $n \geq 1$, we have

$$B_n = \sum_{k=0}^{n-1} \binom{n-1}{k} B_k.$$

Hint: There is a block containing 1. Consider cases based on how many elements this block contains. Let k be the number of elements that are not in the block containing 1.

2.5 The Binomial Theorem

Recall that $\binom{n}{k}$ counts the number of subsets of size k taken from a set of size n. Why are these numbers called binomial coefficients? In general a binomial is just a polynomial with two terms. Let's explore a bit. We see that

$$\begin{aligned}(x+y)^2 &= (x+y)(x+y)\\ &= xx + xy + yx + yy\\ &= x^2 + 2xy + y^2.\end{aligned}$$

The coefficients in this expansion are $1, 2, 1$. Hey, we saw those in row $n = 2$ of Pascal's Triangle! Let's try a larger example. We see that

$$\begin{aligned}(x+y)^3 &= (x+y)^2(x+y)\\ &= (xx + xy + yx + yy)(x+y)\\ &= xxx + xyx + yxx + yyx + xxy + xyy + yxy + yyy\\ &= x^3 + 3x^2y + 3xy^2 + y^3.\end{aligned}$$

This time the coefficients are $1, 3, 3, 1$, which is the next row of Pascal's Triangle. What's going on here?

Consider the expansion of $(x+y)^n$. The key observation is that *before* commuting factors and collecting like terms, the terms in the expansion consist of *all possible* ordered products (i.e., words) where we choose either x or y from each factor. Each such term will consist of n letters. In particular, if there are k copies of x in a term, there will be $n-k$ copies of y. Moreover, every possible configuration of k copies of x (i.e., location of the x's in the term *before* doing any commuting with x and y) will be represented. This means there will be precisely $\binom{n}{k}$ many terms with k copies of x (and $n-k$ copies of y). Thus, when we commute and then collect like terms, the coefficient on $x^k y^{n-k}$ will indeed be $\binom{n}{k}$. This leads us to the following remarkable fact, known as the **Binomial Theorem**.

Theorem 2.42 (Binomial Theorem)**.** For $n \in \mathbb{N} \cup \{0\}$,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

Problem 2.43. Reprove that claim in Problem 2.31 using the Binomial Theorem.

It follows from the Binomial Theorem that

$$(1+t)^n = \sum_{k=0}^{n} \binom{n}{k} t^k = \binom{n}{0} t^0 + \binom{n}{1} t^1 + \cdots + \binom{n}{n} t^n.$$

That is, the expression $(1+t)^n$ encodes the entries in the nth row of Pascal's Triangle. In particular, $\binom{n}{k}$ is the coefficient on t^k in the expansion of $(1+t)^n$. This is our first example of what is called a **generating function**.

Chapter 3

Generating Functions

3.1 Ordinary Generating Functions

The (**ordinary**) **generating function** (or **generating series**) for a sequence $a_0, a_1, a_2, \ldots$ is the formal series

$$\sum_{k\geq 0} a_k t^k = a_0 + a_1 t^1 + a_2 t^2 + \cdots$$

In this context, we treat t as an indeterminate. Two formal power series are equal precisely when their corresponding coefficients are equal. Under most circumstances, we will not be concerned with issues related to convergence of the series. For a finite sequence, we take all subsequent terms to be zero.

A generating function is an algebraic device for encoding a sequence. Herbert Wilf wrote,

> *A generating function is a clothesline on which we hang up a sequence of numbers for display.*

We define operations on formal power series to agree with the usual operations on polynomials. Below, we define scalar multiplication, addition, and multiplication, respectively. In each case, including the product, each coefficient on the right-hand side involves only finitely many terms, so these operations require no assumptions about convergence.

Definition 3.1. Let $a_0, a_1, a_2, \ldots$ and $b_0, b_1, b_2, \ldots$ be two sequences of numbers and let c be a constant. Then:

(a) $c \sum_{k\geq 0} a_k t^k = \sum_{k\geq 0} c a_k t^k$;

(b) $\left(\sum_{k\geq 0} a_k t^k\right) + \left(\sum_{k\geq 0} b_k t^k\right) = \sum_{k\geq 0} (a_k + b_k) t^k$;

(c) $\left(\sum_{k\geq 0} a_k t^k\right) \cdot \left(\sum_{\ell\geq 0} b_\ell t^\ell\right) = \sum_{j\geq 0} \left(\sum_{k+\ell=j} a_k b_\ell\right) t^j = \sum_{j\geq 0} \left(\sum_{k=0}^{j} a_k b_{j-k}\right) t^j$.

Figure 3.1 illustrates the multiplication in Definition 3.1(c): terms on the same diagonal have the same total degree $j = k + \ell$. Summing along that diagonal gives the coefficient $\sum_{k=0}^{j} a_k b_{j-k}$ of t^j.

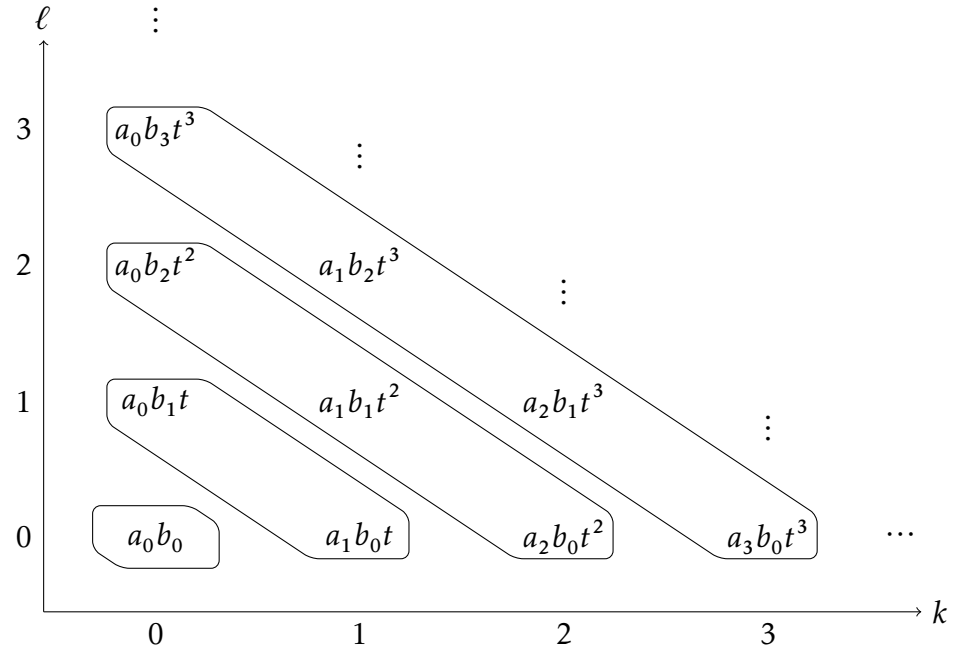


Figure 3.1: Multiplying formal power series by collecting like terms. The entry at (k, ℓ) is $a_k b_\ell t^{k+\ell}$; the outlined groups collect the terms of degrees 0, 1, 2, and 3.

Example 3.2. Here are two examples of generating functions.

(a) The **geometric series** satisfies

$$1 + t + t^2 + \cdots = \sum_{k \geq 0} t^k = \frac{1}{1-t}.$$

Each expression above encodes the sequence $1, 1, 1, \ldots$. The expression $\frac{1}{1-t}$ on the far right is a **closed form** for the generating function. In the context of calculus, you may remember that the equality above only holds for $|t| < 1$. However, in our formal setting, each expression in the equation above should be interpreted as the multiplicative inverse of $1 - t$. Indeed, we have

$$\begin{aligned}(1-t)(1+t+t^2+\cdots) &= (1+t+t^2+\cdots) - t(1+t+t^2+\cdots) \\ &= (1+t+t^2+\cdots) - (t+t^2+\cdots) \\ &= 1.\end{aligned}$$

(b) For $n \in \mathbb{N} \cup \{0\}$, the Binomial Theorem tells us that

$$\binom{n}{0}t^0 + \binom{n}{1}t^1 + \cdots + \binom{n}{n}t^n = \sum_{k=0}^{n}\binom{n}{k}t^k = (1+t)^n.$$

For a fixed n, this is the generating function for the finite sequence $\binom{n}{0}, \binom{n}{1}, \ldots, \binom{n}{n}$. The expression $(1+t)^n$ on the far right is a closed form for the generating function.

Problem 3.3. What sequence is $F(t) = \dfrac{1}{1+t}$ the generating function for?

Problem 3.4. What sequence is $F(t) = \dfrac{1}{t-3}$ the generating function for?

Problem 3.5. What sequence is $F(t) = \dfrac{1}{(1-t)^2}$ the generating function for?

Problem 3.6. Find a closed form for the generating function for the sequence $0, 1, 2, 3, \ldots$.

The next problem tells us that given the generating series for a sequence of numbers, we can obtain the generating series for the sequence of partial sums by simply multiplying by the geometric series.

Problem 3.7. Prove that if $F(t) = \sum_{k\geq 0} a_k t^k$, then

$$\sum_{k\geq 0}(a_0 + a_1 + \cdots + a_k)t^k = \frac{F(t)}{1-t}.$$

Problem 3.8. Explain why Problem 3.5 is a special case of Problem 3.7.

Recall that for $n \in \mathbb{N}$, the nth **triangular number** is defined via

$$T_n := 1 + 2 + \cdots + n$$

and we also set $T_0 = 0$ by convention.

Problem 3.9. Find a closed form for the generating function for the triangular numbers such that T_n is the coefficient on t^n.

Problem 3.10. What sequence is

$$F(t) = \frac{1}{1 - 5t + 6t^2}$$

the generating function for?

Hint: One approach is to use a partial fraction decomposition. Alternatively, you could use the formula for the product of two sums together with the following fact that you may freely use:

$$(x-y)(x^k + x^{k-1}y + x^{k-2}y^2 + \cdots + xy^{k-1} + y^k) = x^{k+1} - y^{k+1}.$$

We define differentiation and integration of generating series coefficientwise, without requiring convergence. For these operations, we assume that the coefficients are real numbers.

Definition 3.11. If $a_0, a_1, a_2, \ldots$ is a sequence of numbers, then we define the following.

(a) $\dfrac{d}{dt}\left[\displaystyle\sum_{k\geq 0} a_k t^k\right] := \sum_{k\geq 1} k a_k t^{k-1} = \sum_{k\geq 0} (k+1) a_{k+1} t^k;$

(b) $\displaystyle\int \sum_{k\geq 0} a_k t^k \, dt := C + \sum_{k\geq 0} \frac{a_k}{k+1} t^{k+1}$, where C is an arbitrary constant;

(c) $\displaystyle\int_0^t \sum_{k\geq 0} a_k x^k \, dx := \sum_{k\geq 0} \frac{a_k}{k+1} t^{k+1}.$

Differentiating any antiderivative of a generating series recovers the original series. In part (b), we obtain a family of generating series, each of which is called an **antiderivative**. In part (c), we obtain the unique antiderivative with constant term zero. Using the definitions above, we can obtain many of the familiar derivative and integral formulas from calculus. For example, if $A(t)$ and $B(t)$ are generating series, then

$$\frac{d}{dt}\big[A(t)B(t)\big] = A'(t)B(t) + A(t)B'(t).$$

The proof involves comparing coefficients of t^k on both sides.

To differentiate a closed form, we can use rules derived from formal differentiation. For example, if $F(t) = P(t)/Q(t)$, where P and Q are polynomials and $Q(0) \neq 0$, then Q has a formal multiplicative inverse. Differentiating $Q(t)F(t) = P(t)$ using the formal product rule yields the usual quotient rule:

$$F'(t) = \frac{P'(t)Q(t) - P(t)Q'(t)}{Q(t)^2}.$$

Problem 3.12. Find a closed form for the generating function for the sequence

$$0, 1, 4, 9, 16, \ldots.$$

Hint: Start with

$$A(t) = \sum_{k\geq 0} kt^k = \frac{t}{(1-t)^2},$$

the generating function for $0, 1, 2, 3, \ldots$. Differentiate $A(t)$, then determine how to shift the powers of t to obtain the desired sequence.

Given a closed form for a generating function, we can always recover the terms of the corresponding sequence by taking successive formal derivatives and extracting their constant terms. This gives a formal version of the familiar Maclaurin expansion from calculus. Here, the closed form is understood to represent a formal power series; evaluating such a series at $t = 0$ means taking its constant term.

Theorem 3.13 (Formal Maclaurin expansion)**.** Let $F(t) = \sum_{k\geq 0} a_k t^k$ be a formal power series with real coefficients. For every integer $k \geq 0$,

$$a_k = \frac{F^{(k)}(0)}{k!},$$

where $F^{(k)}$ denotes the kth formal derivative and $F^{(0)} = F$. Consequently,

$$F(t) = \sum_{k\geq 0} \frac{F^{(k)}(0)}{k!} t^k.$$

Proof. For each integer $k \geq 0$, repeated application of the definition of formal differentiation gives

$$F^{(k)}(t) = \sum_{j\geq k} \frac{j!}{(j-k)!} a_j t^{j-k}.$$

The constant term occurs when $j = k$ and is $k!a_k$. Thus $F^{(k)}(0) = k!a_k$, proving the coefficient formula and hence the series identity. □

Every formal power series can be differentiated arbitrarily many times by definition, so this theorem requires neither convergence nor Taylor's Theorem.

If we instead interpret the closed form as a function of a real variable, convergence becomes a separate issue. An analytic function near 0 equals its Maclaurin series in some neighborhood of 0; rational functions with denominator nonzero at 0 have this property. Infinite differentiability alone does not guarantee equality with the Maclaurin series: to obtain that equality from Taylor's Theorem, the Taylor remainders must tend to zero. The good news is that we do not really need to worry about these details.

In the next two problems, use the familiar derivatives from calculus to find the Maclaurin series, then regard these expansions as formal generating series.

Problem 3.14. Find the Maclaurin series for $F(t) = e^t$. What sequence is $F(t)$ the generating function for?

Problem 3.15. Find the Maclaurin series for $F(t) = \sin(t)$. What sequence is $F(t)$ the generating function for?

Problem 3.16. Use formal integration to find a closed form for the ordinary generating function for the sequence $1, \frac{1}{2}, \frac{1}{6}, \frac{1}{24}, \frac{1}{120}, \ldots$, whose kth term is $a_k = \frac{1}{(k+1)!}$ for $k \geq 0$.

Hint: Integrate the formal exponential series from 0 to t, then shift the powers of t so that the constant term is 1. Before shifting down by one power of t, check that the series has constant term zero. In this situation, division by t means lowering each exponent by one, rather than multiplying by a formal power-series inverse of t, which does not exist.

Problem 3.17. How many ways can you make exactly one dollar using pennies, nickels, dimes, and quarters? Assume that you have an unlimited supply of each denomination, with values 1, 5, 10, and 25 cents, respectively. Coins of the same denomination are indistinguishable, and the order of the coins does not matter. For each integer $k \geq 0$, let a_k be the number of ways to make k cents using these denominations, and define

$$C(t) = \sum_{k \geq 0} a_k t^k.$$

(a) For a coin worth d cents, write a generating series that records the possible total values obtained using only that denomination. Find a closed form for this series.

(b) Use your answer to part (a) to express $C(t)$ as a product of four generating series, and then find a closed form. Explain why the coefficient of t^k counts the desired collections of coins.

(c) Using technology, find the coefficient a_{100} of t^{100} in the corresponding Maclaurin series. How many ways are there to make a dollar using pennies, nickels, dimes, and quarters? In WolframAlpha, you can use the following command:

```
SeriesCoefficient[C(t),{t,0,100}].
```

Let's close out the section with two problems. The first reviews earlier counting techniques, and the second develops a generating function for multisets.

Define a **multiset** M to be an unordered collection of elements that may be repeated. In this section, all multisets are finite. To distinguish between sets and multisets, we will use the notation {} versus {{}}. For example, $M = \{\{a, a, a, b, c, c\}\}$ is a multiset (not to be confused with the set containing the set $\{a, b, c\}$; mathematical notation is both awesome and frustrating at times). The primes occurring in the prime factorization of a natural number greater than 1 is an example of multiset. The cardinality of a multiset is its number of elements counted with multiplicity. So, in our example, $|M| = 3 + 1 + 2 = 6$. If S is a set, then M is a **multiset on** S if every element of M is an element of S. For integers $n, k \geq 0$, with $[0] = \varnothing$, define

$$\left(\!\binom{n}{k}\!\right) := \text{number of multisets on } [n] \text{ of size } k.$$

The only multiset on the empty set is the empty multiset, which has size zero. Thus,

$$\left(\!\binom{0}{0}\!\right) = 1 \qquad \text{and} \qquad \left(\!\binom{0}{k}\!\right) = 0 \quad \text{for } k \geq 1.$$

Problem 3.18. For integers $n \geq 1$ and $k \geq 0$, prove that

$$\left(\!\binom{n}{k}\!\right) = \binom{n+k-1}{k}.$$

Hint: Here is one possible approach. Construct a bijection from the collection of multisets on $[n]$ of size k to the collection of subsets of $[n+k-1]$ of size k. To do this, assume your multisets on $[n]$ are canonically written in nondecreasing order, say $\{\{m_1, m_2, \ldots, m_k\}\}$, where $m_1 \leq m_2 \leq \cdots \leq m_k$. Try to map each such multiset to a k-subset of $[n+k-1]$.

Problem 3.19. For $n \in \mathbb{N} \cup \{0\}$, let $M_n(t)$ denote the generating function for the number of multisets of $[n]$ of size k:

$$M_n(t) := \sum_{k \geq 0} \left(\!\binom{n}{k}\!\right) t^k.$$

Also, define $\mathcal{M}_n := \{A \mid A \text{ is a multiset on } [n]\}$.

(a) Explain why $M_n(t) = \displaystyle\sum_{A \in \mathcal{M}_n} t^{|A|}$, where $|A|$ is the size of the multiset A.

(b) Explain why $\displaystyle\sum_{A \in \mathcal{M}_n} t^{|A|} = (1 + t + t^2 + \cdots)^n$.

(c) Explain why $M_n(t) = \dfrac{1}{(1-t)^n}$.

Hint: For part (b), notice that if you were to expand $(1 + t + t^2 + \cdots)^n$, each resulting term (prior to collecting like terms) corresponds to choosing a term from each of the n factors and then multiplying them together. Each such product of choices corresponds to a unique multiset. In particular, think of the ith factor of $(1+t+t^2+\cdots)^n$ as corresponding to $i \in [n]$. Choosing t^j in the ith factor corresponds to having i occur with multiplicity j in a multiset. When $n = 0$, use the convention that a product with no factors is equal to 1.

3.2 Exponential Generating Functions and Species

The **exponential generating function** (or **exponential generating series**) for a sequence $a_0, a_1, a_2, \ldots$ is the formal series

$$\sum_{k \geq 0} a_k \frac{t^k}{k!} = a_0 \frac{t^0}{0!} + a_1 \frac{t^1}{1!} + a_2 \frac{t^2}{2!} + \cdots.$$

Problem 3.20. For each function below, determine the sequence for which it is the exponential generating function.

(a) $F(t) = \dfrac{1}{1-t}$

(b) $F(t) = e^t$

(c) $F(t) = \sin(t)$

Problem 3.21. Determine all sequences whose exponential generating function $F(t)$ satisfies $F'(t) = F(t)$. How does the answer change if we swap out "exponential" with "ordinary"?

Problem 3.22. Let $A(t) = \displaystyle\sum_{k\geq 0} a_k \frac{t^k}{k!}$ and $B(t) = \displaystyle\sum_{k\geq 0} b_k \frac{t^k}{k!}$. Prove that

$$A(t)B(t) = \sum_{n\geq 0} \left(\sum_{k=0}^{n} \binom{n}{k} a_k b_{n-k} \right) \frac{t^n}{n!}.$$

Given a finite set U, let F be a rule for producing a finite collection of labeled combinatorial structures on U, called F-**structures**. Let $F[U]$ denote the collection of F-structures on the set U. We assume that the number of F-structures on U is uniquely determined by $|U|$. We refer to F as a **combinatorial species** (or just **species** for short). For $n \in \mathbb{N} \cup \{0\}$, we define $F[n] := F[[n]]$ so that we do not need to write two pairs of square brackets. Note that $F[0] = F[\varnothing]$ and for $n \geq 1$, $F[n] = F[\{1, \ldots, n\}]$.

Here is a helpful analogy. Think of a species F as a vending machine that eats a finite set U. We hold out a bag to collect all the F-structures the machine can make out of U. The contents of our bag form the set $F[U]$. What happens when we feed the machine the empty set? For some species, the machine returns a single empty structure. If we represent that structure by $\varnothing$, our bag equals $\{\varnothing\}$, where the outer set braces represent the bag and the inner $\varnothing$ represents the empty structure. In this case, the bag contains one object, even though that object is empty. For other species, the machine returns nothing at all, so our bag is empty: $\{\} = \varnothing$. An empty structure in the bag is different from an empty bag! These are two possibilities; a species could also have more than one structure on the empty set.

Note: The phrase "combinatorial structures" is sufficiently vague here, but for our purposes, we will rely on our intuition. Some examples of labeled combinatorial structures are sets, subsets, permutations, graphs, planar binary trees, partitions, etc.

For $n \in \mathbb{N} \cup \{0\}$, the exponential generating function for the number of F-structures on $[n]$ is given by

$$F(t) := \sum_{n\geq 0} |F[n]| \frac{t^n}{n!}.$$

We say that $F(t)$ is the exponential generating function for the species F.

Using "F" in multiple ways (e.g., species F, F-structures $F[U]$, and exponential generating function $F(t)$) is an example of "overloading the operator", but this is common practice across mathematics. I typically object to using $F(t)$ instead of F to refer to the name of a function. However, I let it slide here due to the potential confusion between a species and its exponential generating function.

There are some details I am leaving out regarding the formal definition of species that you are more than welcome to look up. I'm fairly certain that leaving out these details will not do any damage. Fingers crossed!

Problem 3.23. Let E denote the **species of sets**. An E-structure on a finite set U is simply the set U itself, so $E[U] = \{U\}$, including when $U = \varnothing$. Find a closed form for the exponential generating function

$$E(t) = \sum_{n \geq 0} |E[n]| \frac{t^n}{n!}.$$

Problem 3.24. Let S denote the **species of permutations**. An S-structure on a finite set U is a bijection from U to itself. That is, $S[U] = \{f : U \to U \mid f \text{ is a bijection}\}$. Find a closed form for the exponential generating function

$$S(t) = \sum_{n \geq 0} |S[n]| \frac{t^n}{n!},$$

where $|S[0]| = 1$ (i.e., the empty permutation is the only permutation we can make out of the empty set).

Problem 3.25. Let E_+ denote the **species of nonempty sets**. An E_+-structure on a finite nonempty set U is simply the set U itself, and there are no E_+-structures on the empty set. Thus,

$$E_+[U] = \begin{cases} \{U\}, & U \neq \varnothing, \\ \varnothing, & U = \varnothing. \end{cases}$$

(a) Determine $|E_+[n]|$ for each $n \in \mathbb{N} \cup \{0\}$. Explain how $E_+[0]$ differs from $E[0]$.

(b) Find a closed form for the exponential generating function

$$E_+(t) = \sum_{n \geq 0} |E_+[n]| \frac{t^n}{n!}.$$

(c) How is $E_+(t)$ related to $E(t)$?

The E_+ vending machine returns nothing when we insert the empty set: $E_+[\varnothing] = \varnothing$, so our bag is empty. In contrast, the E machine returns the empty set itself, so our bag contains one structure: $E[\varnothing] = \{\varnothing\}$. The previous problem generalizes to an arbitrary species.

Problem 3.26. Let F be any species. We define the species F_+ by restricting F to nonempty label sets. That is, for any finite set U,

$$F_+[U] = \begin{cases} F[U], & U \neq \varnothing, \\ \varnothing, & U = \varnothing. \end{cases}$$

Thus, an F_+-structure on a nonempty set is an F-structure on that set, and there are no F_+-structures on the empty set. In the vending-machine analogy, the F_+ machine behaves just like the F machine on nonempty inputs, but returns nothing when we insert the empty set, regardless of what the F machine would return.

(a) Express $F_+(t)$ in terms of $F(t)$ and $|F[0]|$. Under what condition is $F_+(t) = F(t) - 1$?

(b) Describe the species S_+ and find a closed form for its exponential generating function.

Problem 3.27. Fix $k \in \mathbb{N} \cup \{0\}$, and let E_k denote the **species of sets of size** k. An E_k-structure on a finite set U is simply the set U itself when $|U| = k$, and there are no E_k-structures on U when $|U| \neq k$. Thus,

$$E_k[U] = \begin{cases} \{U\}, & |U| = k, \\ \varnothing, & |U| \neq k. \end{cases}$$

(a) Determine $|E_k[n]|$ for each $n \in \mathbb{N} \cup \{0\}$. Include the case $k = 0$.

(b) Find a closed form for the exponential generating function

$$E_k(t) = \sum_{n \geq 0} |E_k[n]| \frac{t^n}{n!}.$$

Problem 3.28. Let U be a finite set with $n = |U|$ elements. We identify a linear order of U with a bijection $f : [n] \to U$, which we may represent using one-line notation: $f(1) \cdots f(n)$. For example, 231 represents a linear order f on $[3]$ given by $1 \mapsto 2$, $2 \mapsto 3$, and $3 \mapsto 1$. Let L denote the **species of linear orders**, so an L-structure on U is a linear order of U. There is one linear order on the empty set, represented by the empty word.

(a) Determine $|L[n]|$ for each $n \in \mathbb{N} \cup \{0\}$.

(b) Find a closed form for the exponential generating function

$$L(t) = \sum_{n \geq 0} |L[n]| \frac{t^n}{n!}.$$

(c) How is $L(t)$ related to $S(t)$?

Let F and G be two species. We define their **sum**, denoted $F + G$, by

$$(F + G)[U] := F[U] \sqcup G[U],$$

where $\sqcup$ denotes **disjoint union**. Informally, an $(F+G)$-structure on U consists of a choice of one of the two species together with a structure of that species on all of U. We retain which species was chosen, so an object that belongs to both $F[U]$ and $G[U]$ contributes two distinct structures to $(F + G)[U]$.

Problem 3.29. Prove that

$$(F + G)(t) = F(t) + G(t),$$

where F and G are two species.

Problem 3.30. Let $k, n \in \mathbb{N} \cup \{0\}$.

(a) Describe $(E_0 + E_1 + \cdots + E_k)[n]$.

(b) Determine $|(E_0 + E_1 + \cdots + E_k)[n]|$.

(c) Describe $E_0 + E_1 + \cdots + E_k$.

(d) Determine $(E_0 + E_1 + \cdots + E_k)(t)$.

Let F and G be two species. We define their **product**, denoted $F \cdot G$, by

$$(F \cdot G)[U] := \bigsqcup_{A \subseteq U} F[A] \times G[U \setminus A],$$

where $\sqcup$ denotes "disjoint union". Informally, an $(F \cdot G)$-structure is an ordered pair formed by an F-structure and a G-structure over complementary disjoint sets. That is, we put an F-structure on a subset of U and then a G-structure on that subset's complement.

Problem 3.31. Prove that

$$(F \cdot G)(t) = F(t)G(t),$$

where F and G are two species.

Problem 3.32. A **bicolored set** is a set in which every element is colored one of two colors (say red or blue). For example, $\{a, b, c\}$ and $\{a, b, c\}$ are two different bicolored sets.

(a) Explain why every collection of bicolored sets with n elements is in bijection with the collection of $(E \cdot E)$-structures on $[n]$, where E is the species introduced in Problem 3.23. It follows that the corresponding exponential generating function is $(E \cdot E)(t) = E(t)E(t) = e^t e^t = e^{2t}$.

(b) To really drive the point home, use Problem 3.22 to compute $E(t)E(t)$.

Problem 3.33. Let P denote the **species of subsets**. That is, a P-structure on $[n]$ is a subset of $[n]$. Find a closed form for $P(t)$.

Let F and G be two species. Further, let's suppose that $G[\varnothing] = \varnothing$ (i.e., there are no G-structures on the empty set). Using F and G, we can construct new structures by "composing" F and G.

We define their **composition**, denoted $F \circ G$, as follows: An $(F \circ G)$-structure on a finite set U is a triplet $s = (\pi, \phi, \gamma)$, where

- π is a partition on U;
- $\gamma = (\gamma_B)_{B \in \pi} \in \prod_{B \in \pi} G[B]$, so γ assigns a G-structure to each block of π;
- ϕ is an F-structure on the set of blocks of π.

If any block admits no G-structures, then no such γ exists, and this partition contributes no $(F \circ G)$-structures (see Problem 2.2).

This looks more complicated than it is. An $(F \circ G)$-structure of U simply means we put G-structures on each of the blocks of a partition of U and then put an F-structure on the collection of blocks.

Thus, for any finite set U,

$$(F \circ G)[U] = \left\{ (\pi, \phi, \gamma) \;\middle|\; \begin{array}{l} \pi \text{ is a partition of } U, \\ \phi \in F[\pi], \\ \gamma = (\gamma_B)_{B \in \pi}, \quad \gamma_B \in G[B] \text{ for every } B \in \pi \end{array} \right\}.$$

By grouping the last two entries of each triple, we obtain a canonical bijection

$$(F \circ G)[U] \longleftrightarrow \bigsqcup_{\pi \text{ partition of } U} F[\pi] \times \prod_{B \in \pi} G[B].$$

Here the disjoint union retains the indexing partition π as a tag, so an element in the component indexed by π is represented by $(\pi, (\phi, \gamma))$. The bijection is given by

$$(\pi, \phi, \gamma) \longmapsto (\pi, (\phi, \gamma)).$$

Remark. We will often describe an $(F \circ G)$-structure using a simpler representation of the tuple (π, ϕ, γ). That is, we may identify the tuple with this representation when each uniquely determines the other. For species, we also require these identifications to respect relabeling: changing the labels before or after passing to the representation gives the same result.

Example. An $(L \circ E_2)$-structure on a finite set U corresponds uniquely to an ordered list of unordered pairs that partition U. Think of forming teams of two and then lining up the teams. The E_2-structures make each block a pair, and the L-structure puts the collection of pairs in a linear order.

For example, on $U = [6]$, take

$$\pi = \big\{\{1, 4\}, \{2, 6\}, \{3, 5\}\big\}.$$

Define the linear order $\phi : [3] \to \pi$ by

$$\phi(1) = \{2, 6\}, \qquad \phi(2) = \{1, 4\}, \qquad \phi(3) = \{3, 5\}.$$

Each block $B \in \pi$ has a unique E_2-structure, namely B itself. Thus, with $\gamma = (\gamma_B)_{B\in\pi}$ defined by $\gamma_B = B$, the formal tuple

$$s = (\pi, \phi, \gamma)$$

represents the ordered list

$$\big(\{2,6\},\{1,4\},\{3,5\}\big).$$

Changing the order of the pairs changes the structure. Writing $\{6,2\}$ instead of $\{2,6\}$ does not. There are

$$|(L \circ E_2)[6]| = \binom{6}{2}\binom{4}{2}\binom{2}{2} = 90$$

such structures on $[6]$: choose the first pair, then the second, then the third.

On $[5]$, the entire collection is

$$(L \circ E_2)[5] = \big\{(\pi, \phi, (\gamma_B)_{B\in\pi}) \mid \pi \text{ partitions } [5],\ \phi \in L[\pi],\ \gamma_B \in E_2[B] \text{ for every } B \in \pi\big\} = \varnothing,$$

since a set with five elements cannot be partitioned into pairs. There are no tuples to list.

Remark. Why do we require $G[\varnothing] = \varnothing$? The blocks of a set partition are nonempty, so composition uses G-structures only on nonempty label sets. In contrast, the species product G^k (the product of k copies of G) allows empty parts. Requiring $G[\varnothing] = \varnothing$ ensures that these empty parts contribute no structures, allowing $G(t)^k$ to count ordered collections of k nonempty blocks carrying G-structures.

Example. Let $\mathcal{G}$ denote the **species of simple graphs**. A $\mathcal{G}$-structure on a finite set U is a graph with vertex set U whose edges are unordered pairs of distinct elements of U. There is one graph on the empty vertex set, namely the empty graph.

An $(E \circ \mathcal{G}_+)$-structure corresponds uniquely to a set partition together with a simple graph on each block. To satisfy the empty-set condition for composition, we use $\mathcal{G}_+$, the species of simple graphs with nonempty vertex sets. These graphs may have no edges. For example, on $[5]$, let

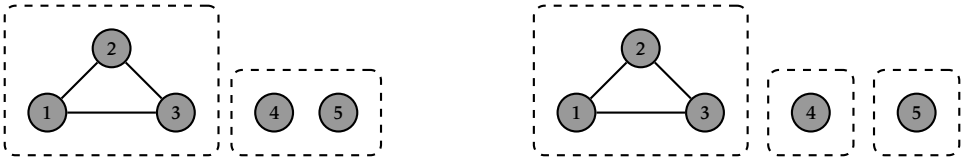
$$\pi = \big\{\{1,2,3\},\{4,5\}\big\}.$$

On the first block, construct the graph with edges $\{1,2\}$, $\{2,3\}$, and $\{1,3\}$, and on the second block, construct the graph with no edges. In the formal tuple $s = (\pi, \phi, \gamma)$, γ records the graph on each block.

The left diagram in Figure 3.2 represents the structure just described. The right diagram represents a different structure, with partition

$$\pi' = \big\{\{1,2,3\},\{4\},\{5\}\big\}.$$

Both give the same graph on $[5]$ when we forget the blocks and take the union of the edges. Hence the graph alone does not determine the $(E \circ \mathcal{G}_+)$-structure: we must retain the partition as well.



Two blocks Three blocks

Figure 3.2: Two distinct $(E \circ \mathcal{G}_+)$-structures on $[5]$ with the same underlying graph. Dashed boundaries indicate the blocks.

Problem 3.34. Prove that

$$(F \circ G)(t) = F(G(t)),$$

where F and G are two species such that $G[\varnothing] = \varnothing$.

Example. Recall the earlier example in which $(L \circ E_2)$-structures correspond uniquely to ordered lists of unordered pairs. Since $E_2[\varnothing] = \varnothing$, Problem 3.34 applies. Using $L(t) = 1/(1-t)$ and $E_2(t) = t^2/2$, we obtain

$$(L \circ E_2)(t) = L(E_2(t)) = \frac{1}{1 - t^2/2}.$$

Expanding as a geometric series and then writing the result as an exponential generating series gives

$$\frac{1}{1 - t^2/2} = \sum_{k \geq 0} \frac{t^{2k}}{2^k} = \sum_{k \geq 0} \frac{(2k)!}{2^k} \frac{t^{2k}}{(2k)!}.$$

Thus, there are $(2k)!/2^k$ structures on $[2k]$ and no structures on a label set of odd size. In particular, this recovers the earlier counts

$$|(L \circ E_2)[6]| = \frac{6!}{2^3} = 90 \qquad \text{and} \qquad |(L \circ E_2)[5]| = 0.$$

The constant term is 1, corresponding to the empty list of pairs on the empty set.

Problem 3.35. Let $\mathcal{C}$ denote the **species of cycles**. For a nonempty finite set U, a $\mathcal{C}$-structure on U is a cyclic ordering of all its elements, with orderings that differ by a cyclic rotation identified. For example,

$$\mathcal{C}[\{a, b, c\}] = \{(a, b, c), (a, c, b)\}.$$

These two cycles are depicted in Figure 3.3. There are no $\mathcal{C}$-structures on the empty set. Recall from abstract algebra that a permutation on a finite set U can be represented in cycle notation. For example, the permutation $w : [6] \to [6]$ with one-line notation $w = 345216$ can be written as the product of disjoint cycles

$$w = (1, 3, 5)(2, 4)(6).$$

Here, $1 \mapsto 3 \mapsto 5 \mapsto 1$, $2 \mapsto 4 \mapsto 2$, and 6 is fixed. We include the singleton cycle (6) to account for every element of $[6]$.

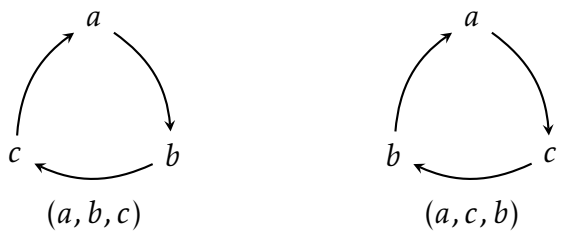


Figure 3.3: The two cycles on $\{a, b, c\}$. Arrows indicate the successor of each element.

(a) Explain why a permutation of U corresponds uniquely to an $(E \circ \mathcal{C})$-structure on U.

(b) Use part (a) and Problem 3.34 to find a closed form for $\mathcal{C}(t)$.

(c) Explain directly why $|\mathcal{C}[n]| = (n-1)!$ for $n \geq 1$.

(d) Expand your answer from part (b) as an exponential generating series and verify that it gives the same count as part (c).

Hint: Compute both sides of

$$\int_0^t \frac{1}{1-x}\,dx = \int_0^t \sum_{n\geq 0} x^n\,dx.$$

Problem 3.36. Recall the definition of set partition given earlier. Let Par denote the **species of set partitions**. Use Problem 3.34 to find a closed form for $\mathrm{Par}(t)$.

Problem 3.37. For $n \in \mathbb{N} \cup \{0\}$, an **ordered set partition** (or **ballot**) of $[n]$ is a set partition of $[n]$ together with a linear order on its collection of blocks. For $n \geq 1$, we represent it by a list $(B_1, B_2, \ldots, B_k)$ containing each block exactly once, where $1 \leq k \leq n$. For $n = 0$, the collection of blocks is empty and has a unique linear order, so the unique ordered set partition is the empty list $()$. Let Bal denote the **species of ordered set partitions**. Find a closed form for $\mathrm{Bal}(t)$.

Problem 3.38. Let $n \in \mathbb{N} \cup \{0\}$. For $n \geq 1$, a **Cayley permutation** on $[n]$ is a function $w : [n] \to [n]$ whose image is $[k]$ for some $1 \leq k \leq n$. We write w in one-line notation as $w(1) \cdots w(n)$. For $n = 0$, we take the empty word to be the unique Cayley permutation on $[0]$. Thus, a Cayley permutation is a word of positive integers, possibly empty, such that whenever b appears, every positive integer $a < b$ also appears.

Let Cay denote the **species of Cayley permutations**. Then $\mathrm{Cay}[n]$ is the collection of Cayley permutations on $[n]$. For example,

$$\mathrm{Cay}[3] = \{111, 112, 121, 122, 123, 132, 211, 212, 213, 221, 231, 312, 321\}.$$

(a) Construct a bijection between $\mathrm{Cay}[n]$ and $\mathrm{Bal}[n]$, the collection of ordered set partitions of $[n]$.

(b) Use this bijection and Problem 3.37 to find a closed form for $\mathrm{Cay}(t)$.

Let F be a species. We define its **derivative species**, denoted F', by

$$F'[U] = F[U \sqcup \{*\}],$$

where $*$ is one additional label not in the finite set U. Thus, an F'-structure on U is an F-structure on the enlarged label set $U \sqcup \{*\}$. The extra label is not counted among the labels of U and remains fixed when we relabel U. In particular, $F'[\varnothing] = F[\{*\}]$.

Problem 3.39. Let F be a species. Prove that

$$F'(t) = \frac{d}{dt}F(t),$$

where $F'(t)$ denotes the exponential generating function of the species F'.

Problem 3.40. Recall that L denotes the species of linear orders. Prove that

$$|(L \cdot L)[U]| = |L'[U]|$$

by describing a bijection between $(L \cdot L)[U]$ and $L'[U]$ for every finite set U, including $U = \varnothing$.

Problem 3.41. By Problems 3.39 and 3.40 and the product formula for species, we have $L'(t) = L(t)^2$. Independently verify this identity by differentiating the closed form for $L(t)$ and comparing the result with its square.

Let F be a species. The **pointed species** associated with F, denoted $F^\bullet$, is defined by

$$F^\bullet[U] = F[U] \times U$$

for each finite set U. An $F^\bullet$-structure is a pair (s, u) consisting of a structure $s \in F[U]$ and a chosen label $u \in U$, called the **distinguished element**. We think of u as the element we are pointing at. Unlike differentiation, pointing selects an existing label rather than adjoining an extra one. In particular, $F^\bullet[\varnothing] = \varnothing$, since there is no label to select.

Example 3.42. There are $3 \cdot 3! = 18$ pointed linear orders on $[3]$. Writing each linear order in one-line notation, we have

$$L^\bullet[3] = \begin{Bmatrix} (123,1), & (123,2), & (123,3), \\ (132,1), & (132,2), & (132,3), \\ (213,1), & (213,2), & (213,3), \\ (231,1), & (231,2), & (231,3), \\ (312,1), & (312,2), & (312,3), \\ (321,1), & (321,2), & (321,3) \end{Bmatrix}.$$

We can represent each pair (w, u) by placing a dot above the designated element u in the word w. Thus, we can simply describe this collection as the set of linear orders on $[3]$ with one designated element, indicated by a dot. In the same order as above, these are

$$\left\{\begin{array}{lll} \dot{1}23, & 1\dot{2}3, & 12\dot{3}, \\ \dot{1}32, & 13\dot{2}, & 1\dot{3}2, \\ \dot{2}13, & 2\dot{1}3, & 21\dot{3}, \\ \dot{2}31, & 23\dot{1}, & 2\dot{3}1, \\ \dot{3}12, & 31\dot{2}, & 3\dot{1}2, \\ \dot{3}21, & 32\dot{1}, & 3\dot{2}1 \end{array}\right\}.$$

Problem 3.43. Recall that $\mathcal{C}$ denotes the species of cycles and L_+ denotes the species of linear orders on nonempty sets.

(a) For every finite set U, describe a bijection between $\mathcal{C}^\bullet[U]$ and $L_+[U]$.

(b) Use your bijection to find a closed form for $\mathcal{C}^\bullet(t)$.

Problem 3.44. Let F be a species. Explain why, for every $n \in \mathbb{N} \cup \{0\}$,

$$|F^\bullet[n]| = n|F[n]|.$$

Problem 3.45. Let F be a species, and recall that E_1 denotes the species of singleton sets. Prove that

$$|F^\bullet[U]| = |(E_1 \cdot F')[U]|$$

by describing a bijection between $F^\bullet[U]$ and $(E_1 \cdot F')[U]$ for every finite set U.

Problem 3.46. Recall that Par denotes the species of set partitions.

(a) For every finite set U, describe a bijection between $\mathrm{Par}^\bullet[U]$ and $(E_1 \cdot E \cdot \mathrm{Par})[U]$.

(b) Use your bijection to find a closed form for $\mathrm{Par}^\bullet(t)$.

Problem 3.47. Let $H = L \circ E_2$, so an H-structure corresponds uniquely to an ordered list of unordered pairs that partition the label set.

(a) For every finite set U, describe a bijection between $H^\bullet[U]$ and $(H \cdot E_2^\bullet \cdot H)[U]$.

(b) Determine $E_2^\bullet(t)$, and use part (a) to find a closed form for $H^\bullet(t)$.

Problem 3.48. Consider the species $\mathrm{Par}^\bullet$ and $E^\bullet \circ E_+$.

(a) Describe the structures of each species on a finite set U. In particular, explain what is distinguished in each case.

(b) For each set partition of $[3]$, determine how many structures of each species have that underlying partition. Use your counts to determine $|\mathrm{Par}^\bullet[3]|$ and $|(E^\bullet \circ E_+)[3]|$.

More coming soon…